```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
from google.colab import files
uploaded = files.upload()
```

Choose Files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving students_20.csv to students_20.csv
```

```
df=pd.read_csv("students_20.csv")
```

Q1. Show the first 5 rows of the DataFrame.

```
print(df.head())
```

```
  Student Name  Math  Science  English
0        Aarav    75      100       86
1         Diya    91       73       93
2        Kabir    94       89       94
3        Meera    88       97       73
4       Ishaan    77       80       74
```

Q2. Show only the "Student Name" and "Math" columns.

```
print(df[["Student Name", "Math"]])
```

```
   Student Name  Math
0         Aarav    75
1          Diya    91
2         Kabir    94
3         Meera    88
4        Ishaan    77
5        Ananya    72
6         Rohan    90
7         Sanya    89
8         Arjun    73
9         Priya    72
10       Vihaan    91
11        Nisha    89
12       Aditya    78
13        Kavya    75
14        Manav    87
15       Ritika    75
16        Karan    80
17       Simran    87
18          Dev    83
19        Pooja    75
```

Q3. Find the average marks in each subject.

```
print(df[["Math", "Science", "English"]].mean())
```

```
Math       82.05
Science    84.60
English    81.35
dtype: float64
```

```
# Calculate mean of selected columns
subject_means = df[["Math", "Science", "English"]].mean()

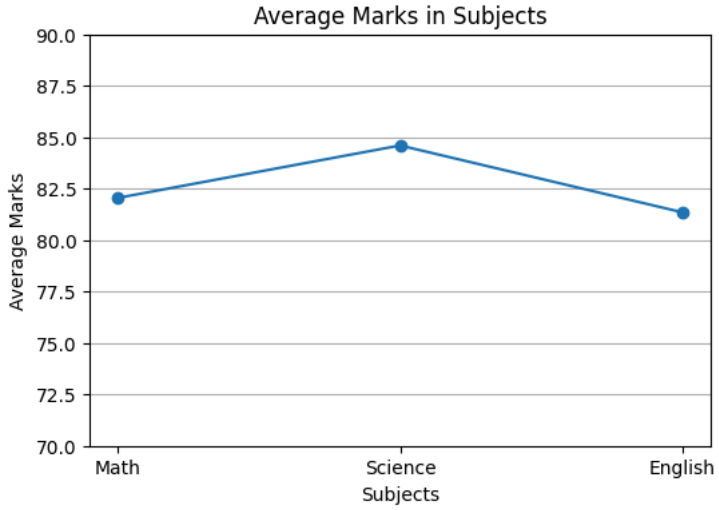
# Plot line graph with grid
plt.figure(figsize=(6,4))
plt.plot(subject_means.index, subject_means.values, marker="o", linestyle="-")

# Labels and title
plt.xlabel("Subjects")
plt.ylabel("Average Marks")
plt.title("Average Marks in Subjects")

plt.grid(axis="y")
```

```
# Set y-axis range
plt.ylim(70, 90)

plt.show()
```



Q4. Which student scored the highest in Science?

Finds the row index where Science marks are maximum.

Fetches the entire row of that student.

```
# Find the maximum Science mark
max_science = df["Science"].max()

# Get the student(s) who scored this maximum
top_science = df[df["Science"] == max_science]

print("Highest Science Marks:", max_science)
print("\nTop Student(s) in Science:\n", top_science)
```

```
Highest Science Marks: 100

Top Student(s) in Science:
   Student Name  Math  Science  English
0        Aarav    75      100       86
```

df["Science"].max() ⟶ gives the highest Science mark.

df[df["Science"] == max_science] ⟶ filters the row(s) where Science = max.

```
max_science = df["Science"].max()
top_students = df.loc[df["Science"] == max_science, "Student Name"]
print(top_students)
```

```
0    Aarav
Name: Student Name, dtype: object
```

df["Science"].max() ⟶ finds the maximum marks in Science (here 100).

df["Science"] == max_science ⟶ creates a Boolean filter for rows with that max value.

.loc[..., "Student Name"] ⟶ selects only the Student Name column for those rows.

```
top_science = df.loc[df["Science"].idxmax(), "Student Name"]
print(top_science)
```

```
Aarav
```

df["Science"].idxmax()

idxmax() returns the index (row number) of the first occurrence of the maximum value in the column.

Example: If the highest Science mark is 95 at row 3, then idxmax() ⟶ 3.

df.loc[...]

.loc[] is used to access rows/columns by labels or index numbers.

So df.loc[3] will return the entire row at index 3.

Q5. Select students who scored more than 90 in English.

```
print(df[df["English"] > 90])
```

```
   Student Name  Math  Science  English
1          Diya    91       73       93
2         Kabir    94       89       94
16        Karan    80       94       97
18          Dev    83       77       91
19        Pooja    75       77       94
```

```
print(df.loc[df["English"] > 90, "Student Name"])
```

```
1      Diya
2     Kabir
16    Karan
18      Dev
19    Pooja
Name: Student Name, dtype: object
```

```
print(df.loc[df["English"] > 90, ["Student Name", "English"]])
```

```
   Student Name  English
1          Diya       93
2         Kabir       94
16        Karan       97
18          Dev       91
19        Pooja       94
```

Q6. Sort students by Math marks in descending order.

```
print(df.sort_values("Math", ascending=False))
```

```
   Student Name  Math  Science  English
2         Kabir    94       89       94
1          Diya    91       73       93
10       Vihaan    91       77       79
6         Rohan    90       81       80
11        Nisha    89       85       70
7         Sanya    89       77       90
3         Meera    88       97       73
17       Simran    87       90       72
14        Manav    87       96       80
18          Dev    83       77       91
16        Karan    80       94       97
12       Aditya    78       94       72
4        Ishaan    77       80       74
0         Aarav    75      100       86
15       Ritika    75       89       75
13        Kavya    75       75       76
19        Pooja    75       77       94
8         Arjun    73       81       84
9         Priya    72       70       75
5        Ananya    72       90       72
```

```
print(df.sort_values("Math", ascending=False)[["Student Name", "Math"]])
```

```
   Student Name  Math
2         Kabir    94
1          Diya    91
10       Vihaan    91
6         Rohan    90
11        Nisha    89
7         Sanya    89
3         Meera    88
17       Simran    87
14        Manav    87
18          Dev    83
16        Karan    80
12       Aditya    78
4        Ishaan    77
0         Aarav    75
15       Ritika    75
13        Kavya    75
19        Pooja    75
8         Arjun    73
9         Priya    72
```

```
5        Ananya    72
```

Q7. Add a new column "Total" = Math + Science + English.

```
df["Total"] = df["Math"] + df["Science"] + df["English"]
print(df)
```

```
   Student Name  Math  Science  English  Total
0         Aarav    75      100       86    261
1          Diya    91       73       93    257
2         Kabir    94       89       94    277
3         Meera    88       97       73    258
4        Ishaan    77       80       74    231
5        Ananya    72       90       72    234
6         Rohan    90       81       80    251
7         Sanya    89       77       90    256
8         Arjun    73       81       84    238
9         Priya    72       70       75    217
10       Vihaan    91       77       79    247
11        Nisha    89       85       70    244
12       Aditya    78       94       72    244
13        Kavya    75       75       76    226
14        Manav    87       96       80    263
15       Ritika    75       89       75    239
16        Karan    80       94       97    271
17       Simran    87       90       72    249
18          Dev    83       77       91    251
19        Pooja    75       77       94    246
```

Q8. Find the student with the highest total marks.

```
top_total = df.loc[df["Total"].idxmax()]
print(top_total)
```

```
Student Name    Kabir
Math               94
Science            89
English            94
Total             277
Name: 2, dtype: object
```

```
top_name = df.loc[df["Total"].idxmax(), "Student Name"]
print(top_name)
```

```
Kabir
```

Q9. Show students who scored less than 80 in at least one subject.

```
print(df[(df["Math"] < 80) | (df["Science"] < 80) | (df["English"] < 80)])
```

```
   Student Name  Math  Science  English  Total
0         Aarav    75      100       86    261
1          Diya    91       73       93    257
3         Meera    88       97       73    258
4        Ishaan    77       80       74    231
5        Ananya    72       90       72    234
7         Sanya    89       77       90    256
8         Arjun    73       81       84    238
9         Priya    72       70       75    217
10       Vihaan    91       77       79    247
11        Nisha    89       85       70    244
12       Aditya    78       94       72    244
13        Kavya    75       75       76    226
15       Ritika    75       89       75    239
17       Simran    87       90       72    249
18          Dev    83       77       91    251
19        Pooja    75       77       94    246
```

Q10. Save the updated DataFrame (with totals) to a new CSV.

```
df.to_csv("students_with_total.csv", index=False)
```

```
df.drop(columns=["Total"]).to_csv("students_without_total.csv", index=False)
```

Setting index=False tells Pandas not to save the index in the file — only your actual data (columns like Student Name, Math, Science, English, Total).

👉 So basically, index=False is used to avoid saving Pandas' index column in the file.

Q11. What is the highest and lowest mark in Math?

```
print("Highest Math:", df["Math"].max())
print("Lowest Math:", df["Math"].min())
```

```
Highest Math: 94
Lowest Math: 72
```

Q12. Find the average total marks of all students.

```
df["Total"] = df["Math"] + df["Science"] + df["English"]
print("Average Total Marks:", df["Total"].mean())
```

```
Average Total Marks: 248.0
```

Q13. Show only the top 3 students in English.

```
print(df.sort_values("English", ascending=False).head(3))
```

```
   Student Name  Math  Science  English  Total
16        Karan    80       94       97    271
2         Kabir    94       89       94    277
19        Pooja    75       77       94    246
```

Q14. Count how many students scored above 85 in Science.

```
print("Count:", (df["Science"] > 85).sum())
```

```
Count: 9
```

Q15. Create a new column "Grade" based on Total Marks.

If Total ≥ 260 ⟶ "A"

If 240 ≤ Total < 260 ⟶ "B"

Else ⟶ "C"

```
df["Grade"] = pd.cut(df["Total"],
                     bins=[0, 239, 259, 300],
                     labels=["C", "B", "A"])
print(df[["Student Name", "Total", "Grade"]])
```

```
   Student Name  Total Grade
0         Aarav    261     A
1          Diya    257     B
2         Kabir    277     A
3         Meera    258     B
4        Ishaan    231     C
5        Ananya    234     C
6         Rohan    251     B
7         Sanya    256     B
8         Arjun    238     C
9         Priya    217     C
10       Vihaan    247     B
11        Nisha    244     B
12       Aditya    244     B
13        Kavya    226     C
14        Manav    263     A
15       Ritika    239     C
16        Karan    271     A
17       Simran    249     B
18          Dev    251     B
19        Pooja    246     B
```

16. Show the average marks per subject for each Grade.

```
# Average marks per subject for each Grade
avg_per_grade = df.groupby("Grade").mean(numeric_only=True)

print(avg_per_grade)
```

```
       Math    Science  English       Total
Grade
C      74.0  80.833333    76.00  230.833333
B      86.1  82.800000    81.40  250.300000
A      84.0  94.750000    89.25  268.000000
/tmp/ipython-input-3056327838.py:2: FutureWarning: The default of observed=False is deprecated and will be changed to True i
  avg_per_grade = df.groupby("Grade").mean(numeric_only=True)
```

```
# Assuming your subject columns are: English, Math, Science
avg_per_grade = df.groupby("Grade")[["English", "Math", "Science"]].mean()

print(avg_per_grade)
```

```
       English  Math    Science
Grade
C        76.00  74.0  80.833333
B        81.40  86.1  82.800000
A        89.25  84.0  94.750000
/tmp/ipython-input-4027114633.py:2: FutureWarning: The default of observed=False is deprecated and will be changed to True i
  avg_per_grade = df.groupby("Grade")[["English", "Math", "Science"]].mean()
```

Q17. Show students who have the same marks in any two subjects.

```
print(df[(df["Math"] == df["Science"]) |(df["Math"] == df["English"]) |(df["Science"] == df["English"])])
```

```
   Student Name  Math  Science  English  Total Grade
2         Kabir    94       89       94    277     A
5        Ananya    72       90       72    234     C
13        Kavya    75       75       76    226     C
15       Ritika    75       89       75    239     C
```

```
same_marks = df[df[["Math", "Science", "English"]].apply(lambda x: x.duplicated().any(), axis=1)]
print(same_marks)
```

```
   Student Name  Math  Science  English  Total Grade
2         Kabir    94       89       94    277     A
5        Ananya    72       90       72    234     C
13        Kavya    75       75       76    226     C
15       Ritika    75       89       75    239     C
```

Q18. Show the top 5 students based on total marks only (Student Name + Total).

```
print(df[["Student Name", "Total"]].sort_values("Total", ascending=False).head(5))
```

```
   Student Name  Total
2         Kabir    277
16        Karan    271
14        Manav    263
0         Aarav    261
3         Meera    258
```

Q19. Save only Student Name and Grade to a new CSV.

```
df[["Student Name", "Grade"]].to_csv("students_grade.csv", index=False)
```

📌 Pandas

Show the first 5 rows of the DataFrame.

Show only the "Student Name" and "Math" columns.

Find the average marks in each subject.

Which student scored the highest in Science?

Select students who scored more than 90 in English.

Sort students by Math marks in descending order.

Add a new column "Total" = Math + Science + English.

Find the student with the highest total marks.

Show students who scored less than 80 in at least one subject.

Save the updated DataFrame (with totals) to a new CSV.

What is the highest and lowest mark in Math?

Find the average total marks of all students.

Show only the top 3 students in English.

Count how many students scored above 85 in Science.

Create a new column "Grade" based on Total Marks.

Show the average marks per subject for each Grade.

Show students who have the same marks in any two subjects.

Show the top 5 students based on total marks only (Student Name + Total).

Save only Student Name and Grade to a new CSV.